Classical versus Deep Learning for Paediatric Pneumonia Screening

Comparative intelligence and ethics in machine learning systems
Machine Learning (PCOM7E), Individual Presentation | University of Essex Online

Why paediatric pneumonia: the errors are not equal

740,180

under-fives killed by pneumonia in 2019

14%

of every under-five death that year

1 in 3

children with pneumonia receive the antibiotics they need

Bacterial pneumonia

Urgent referral for antibiotics

A missed case delays treatment

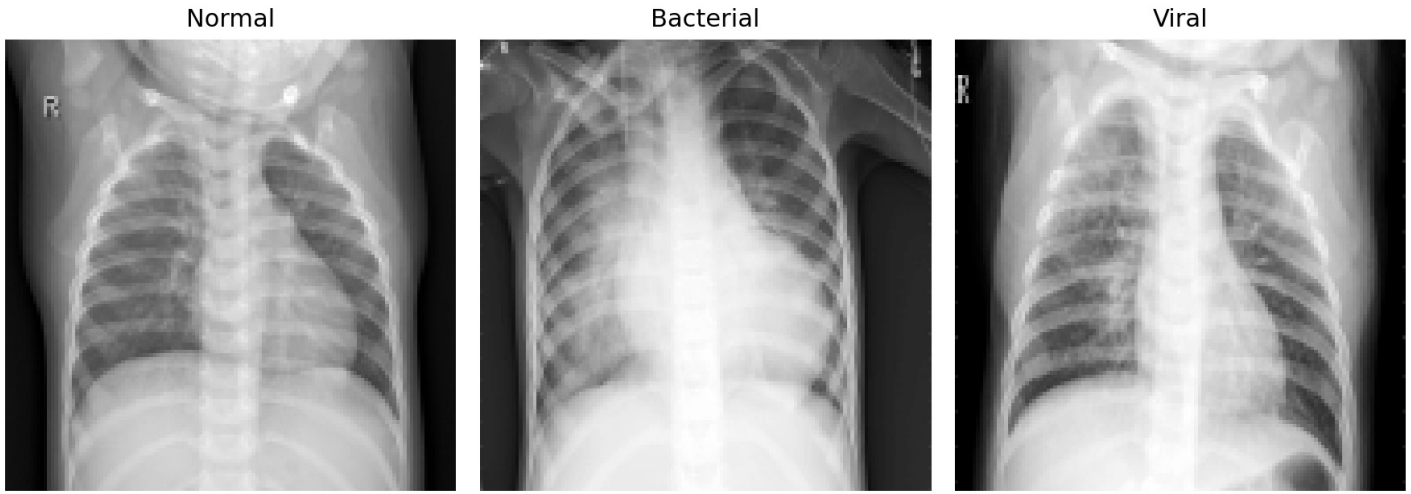
Viral pneumonia

Supportive care

A false alarm costs a second look

World Health Organization (2022); Kermany et al. (2018)

The dataset: 5,856 paediatric chest radiographs



Normal, bacterial and viral examples, resized to 128 x 128

Split	Normal	Bacterial	Viral	Total
Train	1,349	2,538	1,345	5,232
Test	234	242	148	624

Provenance

Ages 1 to 5, Guangzhou, routine care

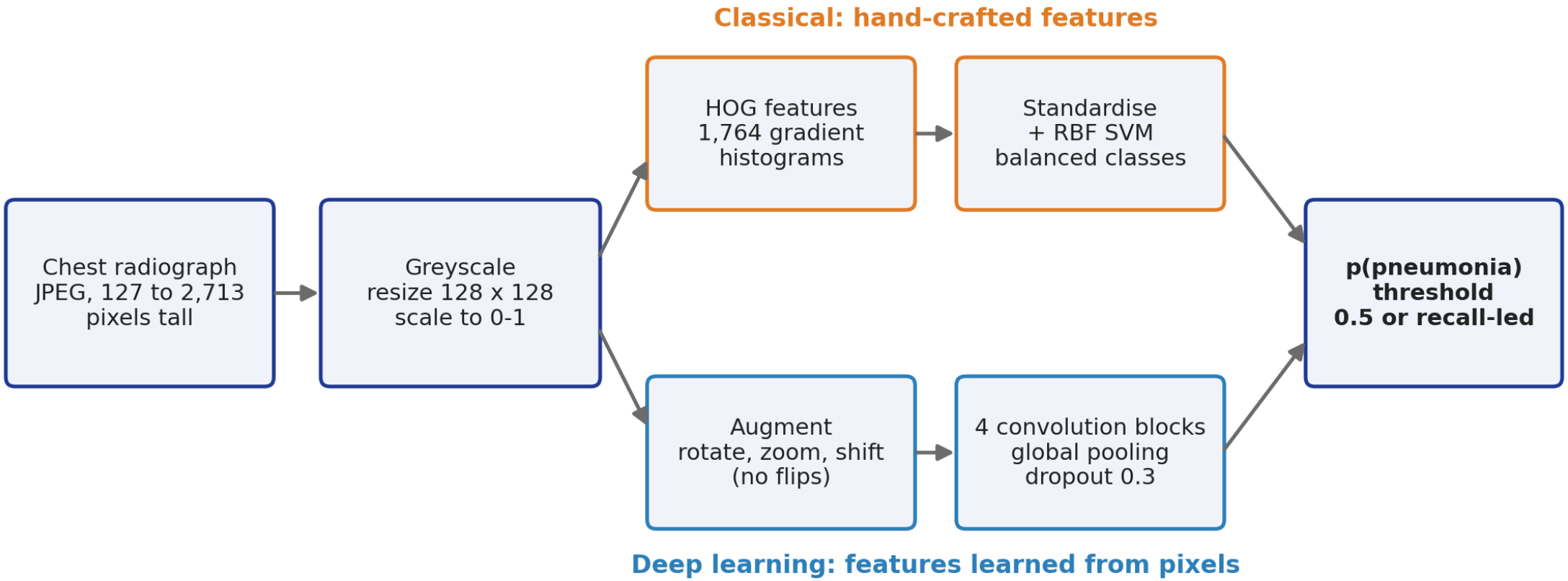
Graded by two physicians

CC BY 4.0, Mendeley Data

Up to 30 images per child; identity only in file names

Kermany et al. (2018); data release: Kermany, Zhang and Goldbaum (2018)

Preprocessing: one input, two representations

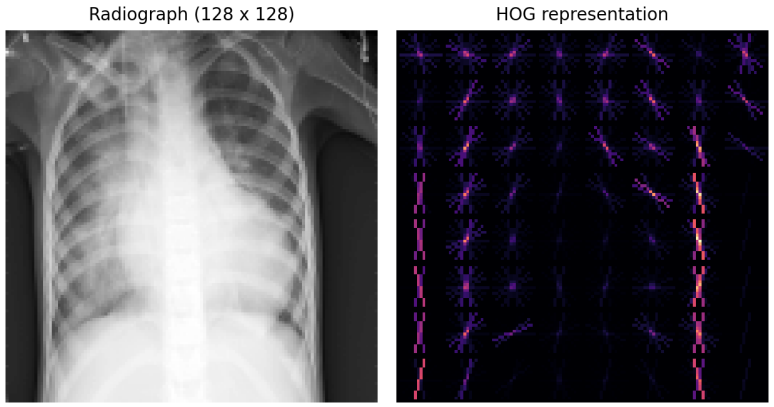


Class weights: normal 1.94, pneumonia 0.67, so each class counts equally in training

HOG descriptor: Dalal and Triggs (2005)

Model design: hand-crafted versus learned features

	HOG + SVM (classical)	CNN (deep learning)
Input	1,764 HOG features	128 x 128 pixels
Model	RBF support vector machine	4 convolution layers, group normalisation, global pooling
Tuning	C by cross-validation; gamma at scale	Adam 0.001, batch 32, early stopping after epoch 6
Training aids	Balanced class weights	Dropout 0.3, rotation, zoom and shift, no flips
Size	1,147 support vectors	240,609 parameters
Library	scikit-learn	TensorFlow / Keras



Fixed in development

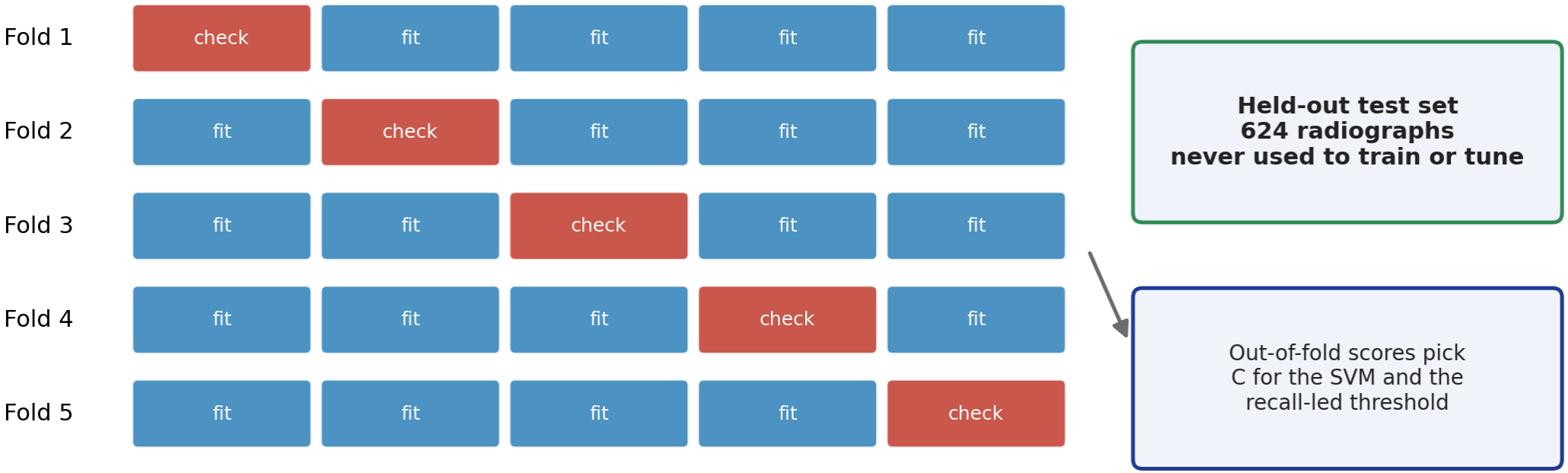
Batch normalisation, then early stopping from epoch 1, each broke two folds

Now: group normalisation, delayed early stopping

Pedregosa et al. (2011)

Validation: five folds, grouped by child

Training set: 5,232 radiographs



Every image from one child stays inside one fold

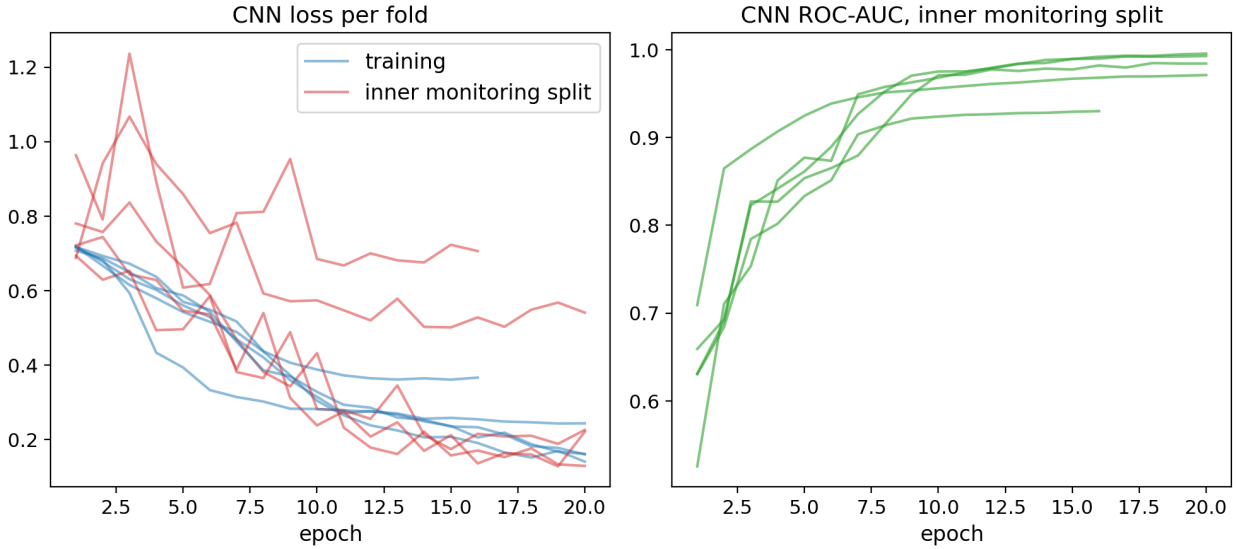
Image-level folds would put 475 to 485 children on both sides of every split; patient-level folds put none

Stratified cross-validation for model selection: Kohavi (1995)

Cross-validation favours the classical model

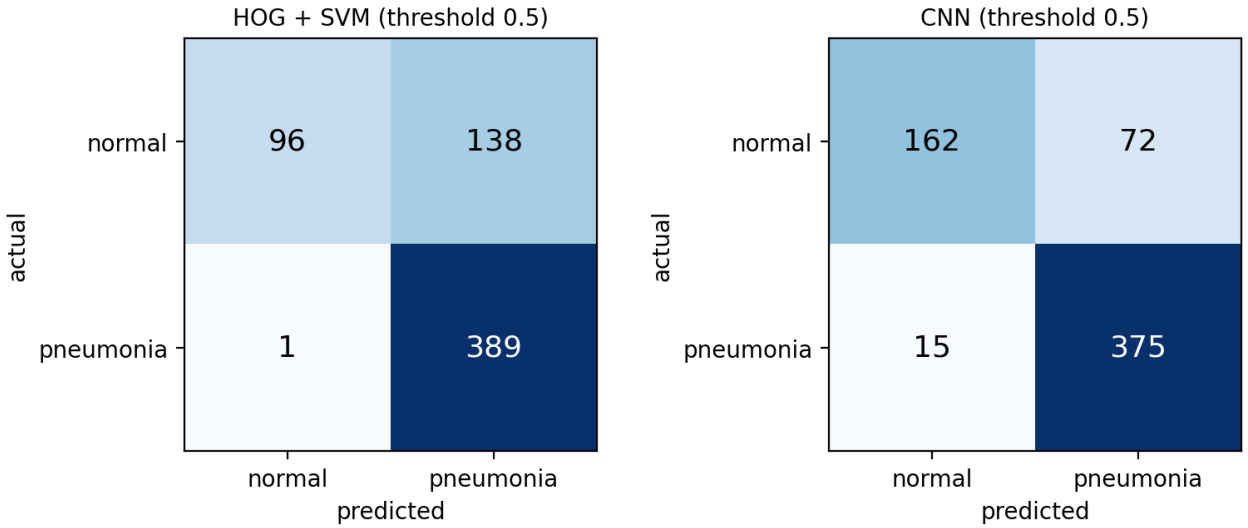
Mean (SD), 5 folds	HOG + SVM	CNN
Accuracy	0.977 (0.005)	0.855 (0.130)
Precision	0.985 (0.004)	0.989 (0.005)
Recall	0.984 (0.004)	0.814 (0.180)
Specificity	0.958 (0.012)	0.973 (0.015)
F1	0.985 (0.003)	0.883 (0.119)
ROC-AUC	0.997 (0.001)	0.973 (0.027)

Suspiciously high: image shape alone reaches 0.834 in cross-validation (slide 13)



CNN folds 3 and 5: ROC-AUC 0.93 and 0.97, yet recall only 0.52 and 0.76 at 0.5

On the held-out test set the ranking reverses



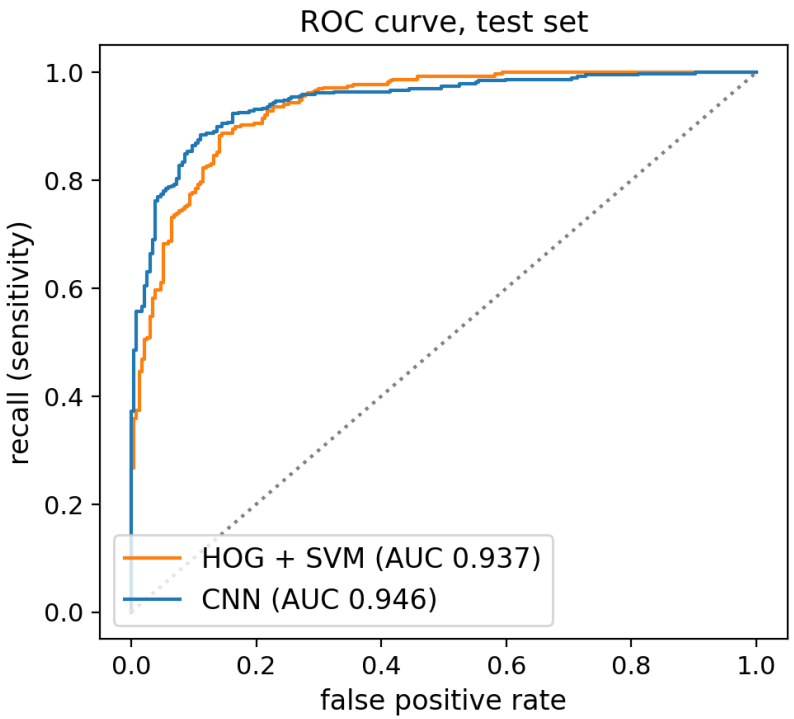
Test set, threshold 0.5	HOG + SVM	CNN
Accuracy	0.777	0.861
Recall	0.997	0.962
Specificity	0.410	0.692
Precision	0.738	0.839
F1	0.848	0.896
ROC-AUC	0.937	0.946

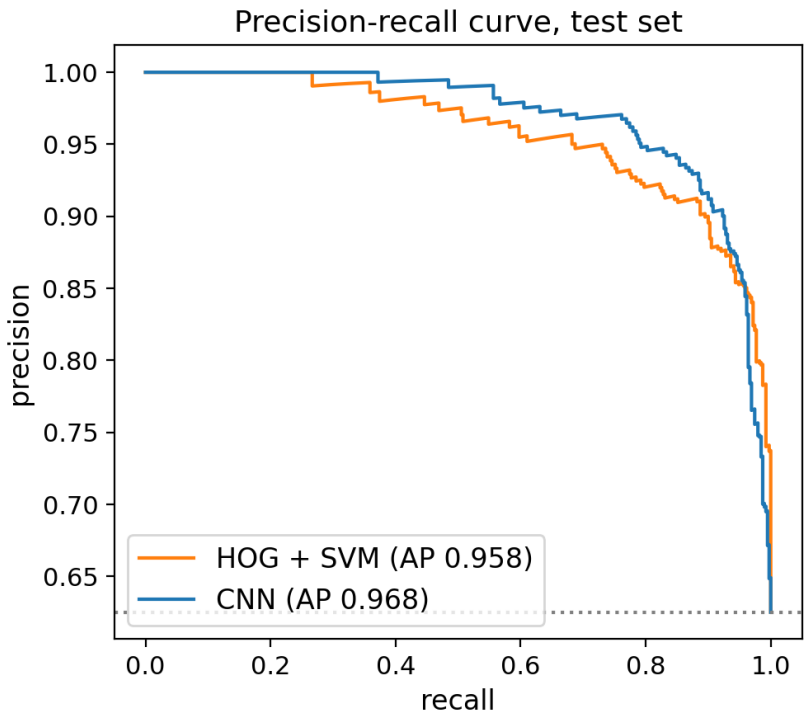
Benchmark: 92.8% accuracy and 90.1% specificity with a pretrained Inception V3 (Kermany et al., 2018)

170 pneumonia identifiers appear in both train and test folders

Specificity: 0.96 and 0.97 in cross-validation, 0.41 and 0.69 on test

Choosing an operating point: recall first



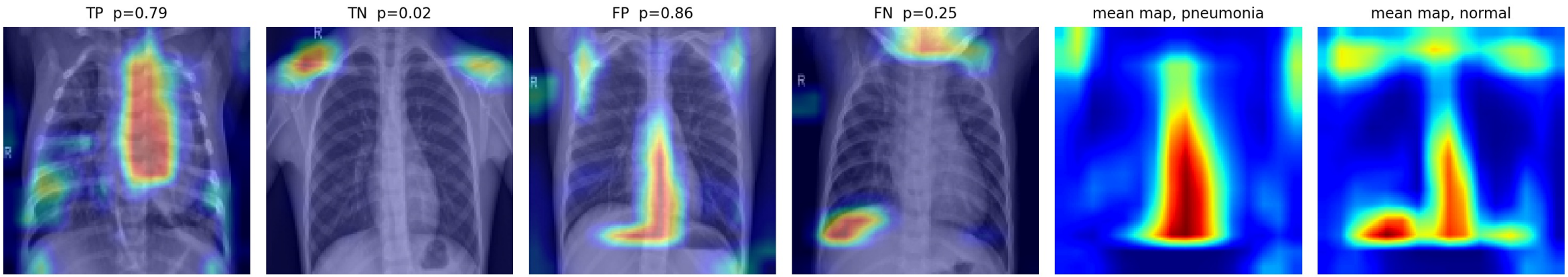


Recall-led threshold	HOG + SVM	CNN
Threshold	0.912	0.114
Recall	0.977	0.987
Specificity	0.641	0.321
Accuracy	0.851	0.737

A cut chosen on training folds shifts on new data

Sculley et al. (2015); Saito and Rehmsmeier (2015)

Grad-CAM: central, but not always clinical



37% / 32%

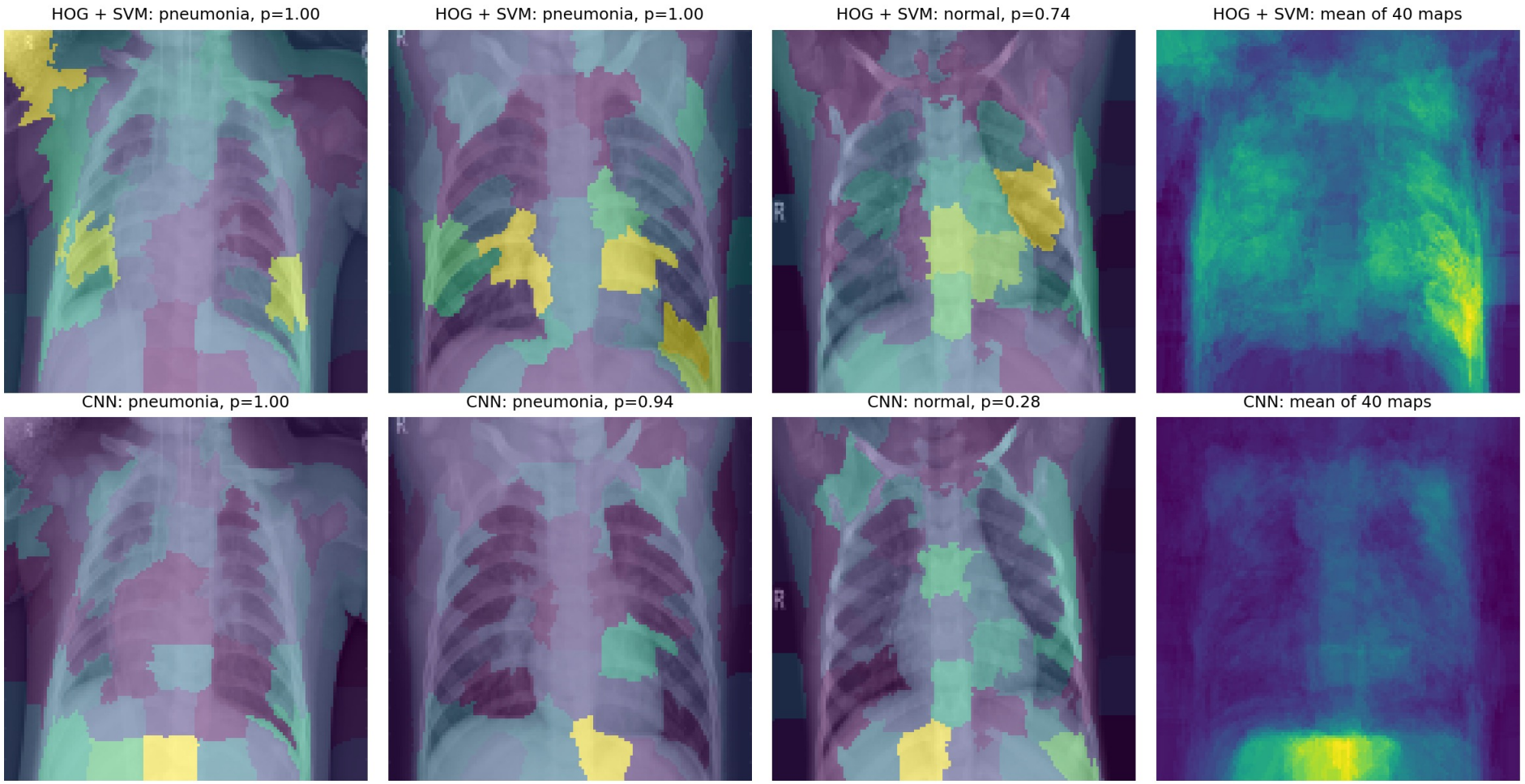
Grad-CAM mass in the border, correct / incorrect calls

44%

border share expected if attention were uniform

Selvaraju et al. (2017)

LIME: both models side by side



36%

of the SVM's importance in the border

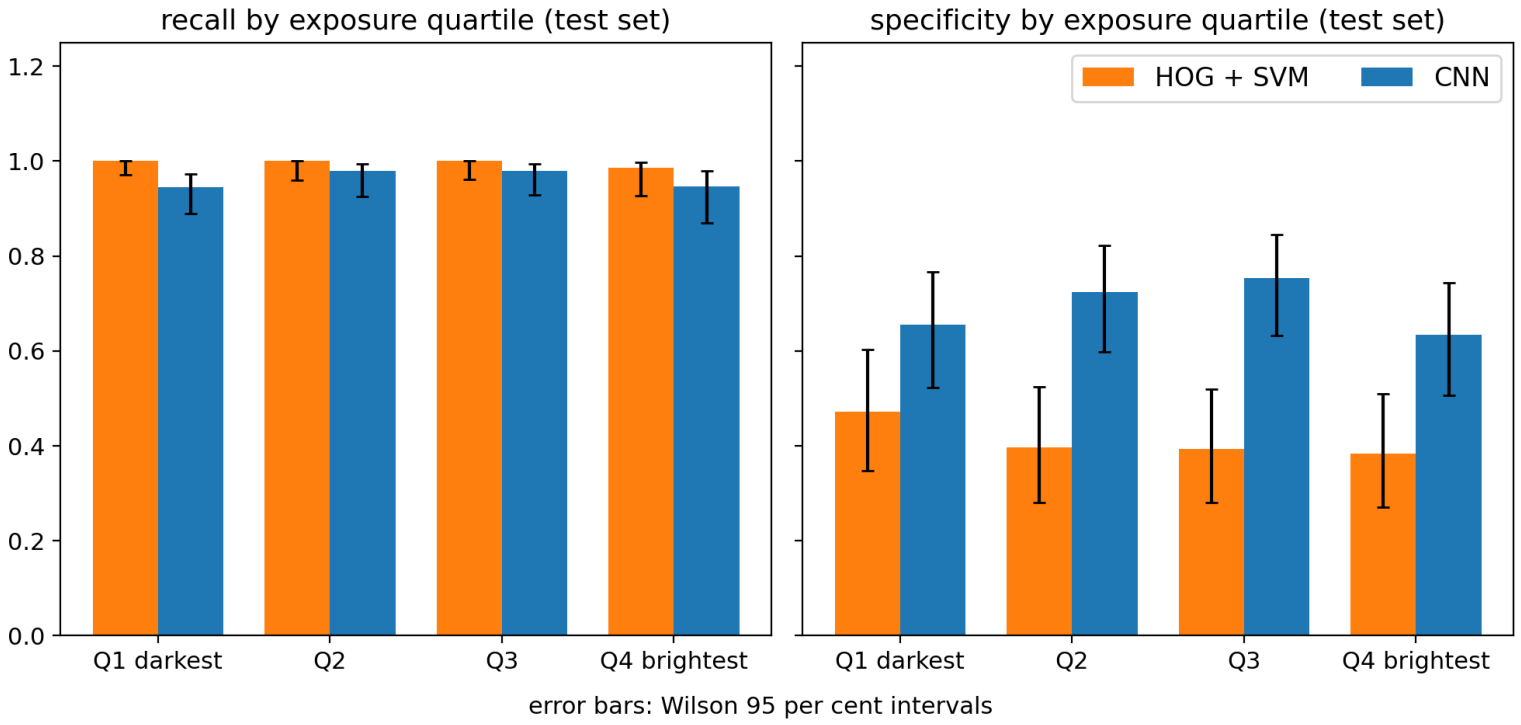
42%

of the CNN's importance in the border

44% expected by area; about 56 regions per image

Ribeiro, Singh and Guestrin (2016)

Subgroups: subtype and exposure



Recall by subtype	HOG + SVM	CNN
Bacterial (242)	1.000	0.971
Viral (148)	0.993	0.946

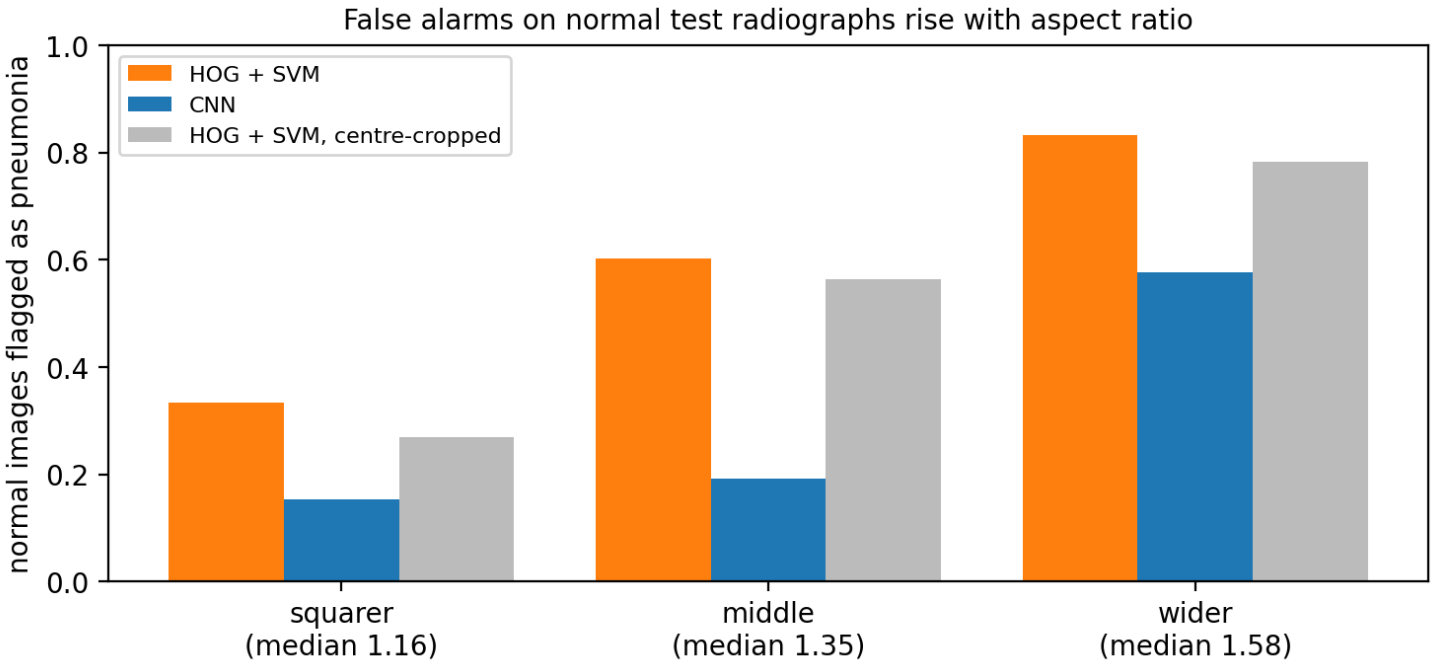
Exposure

SVM specificity 0.47 darkest, 0.38 brightest

Intervals overlap: monitor, not proven

No sex, ethnicity or income recorded

An acquisition shortcut in the data



0.829

test accuracy (0.834 in cross-validation) from image shape alone: above the SVM (0.777), near the CNN (0.861)

Training median	Normal	Pneumonia
Height (px)	1,325	776
Aspect ratio	1.22	1.49

Centre-cropping did not remove it

Shortcut learning: Brown et al. (2023); hospital-system confounding: Zech et al. (2018)

Ethical considerations, grounded in this use case

Privacy and provenance

Routine care, ethics approval

Consent not described

Numeric patient code only; some burned-in text

Licence permits reuse, not consent

Unequal errors

A miss can delay antibiotics

15 of 390

pneumonia radiographs missed by the CNN at 0.5

72 of 234

normal radiographs flagged by the CNN at 0.5

Bias and harm

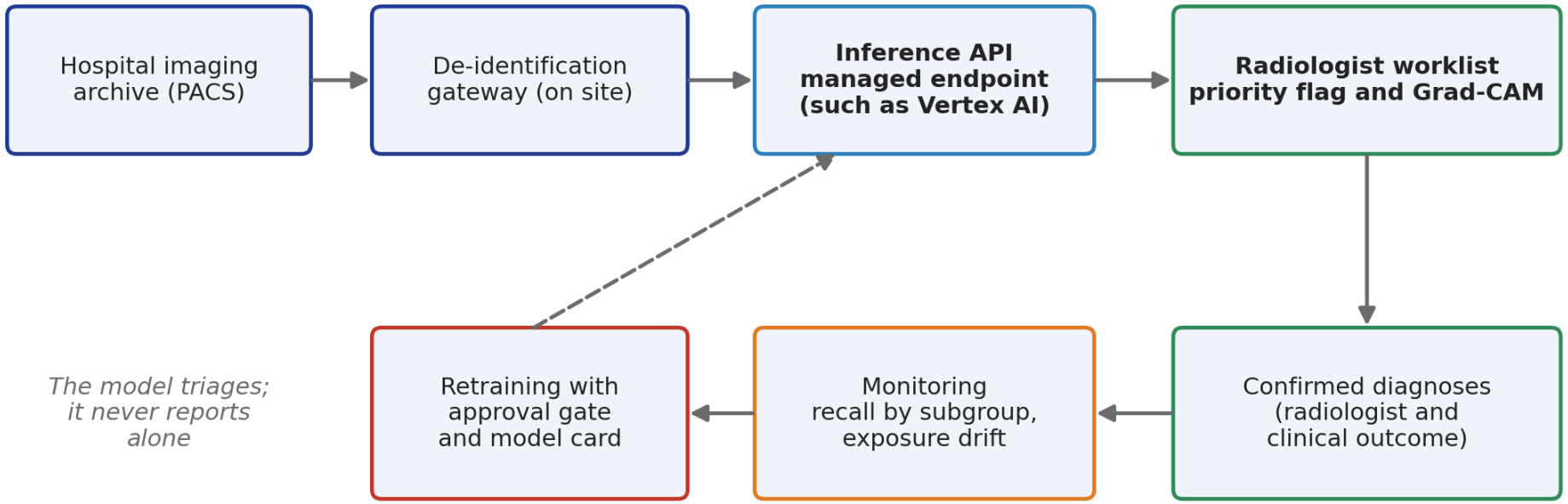
One hospital, one age band

No demographics to audit

Underdiagnosis falls on under-served groups

Seyyed-Kalantari et al. (2021); Kermany et al. (2018)

Recommendation: the CNN, as a monitored triage aid

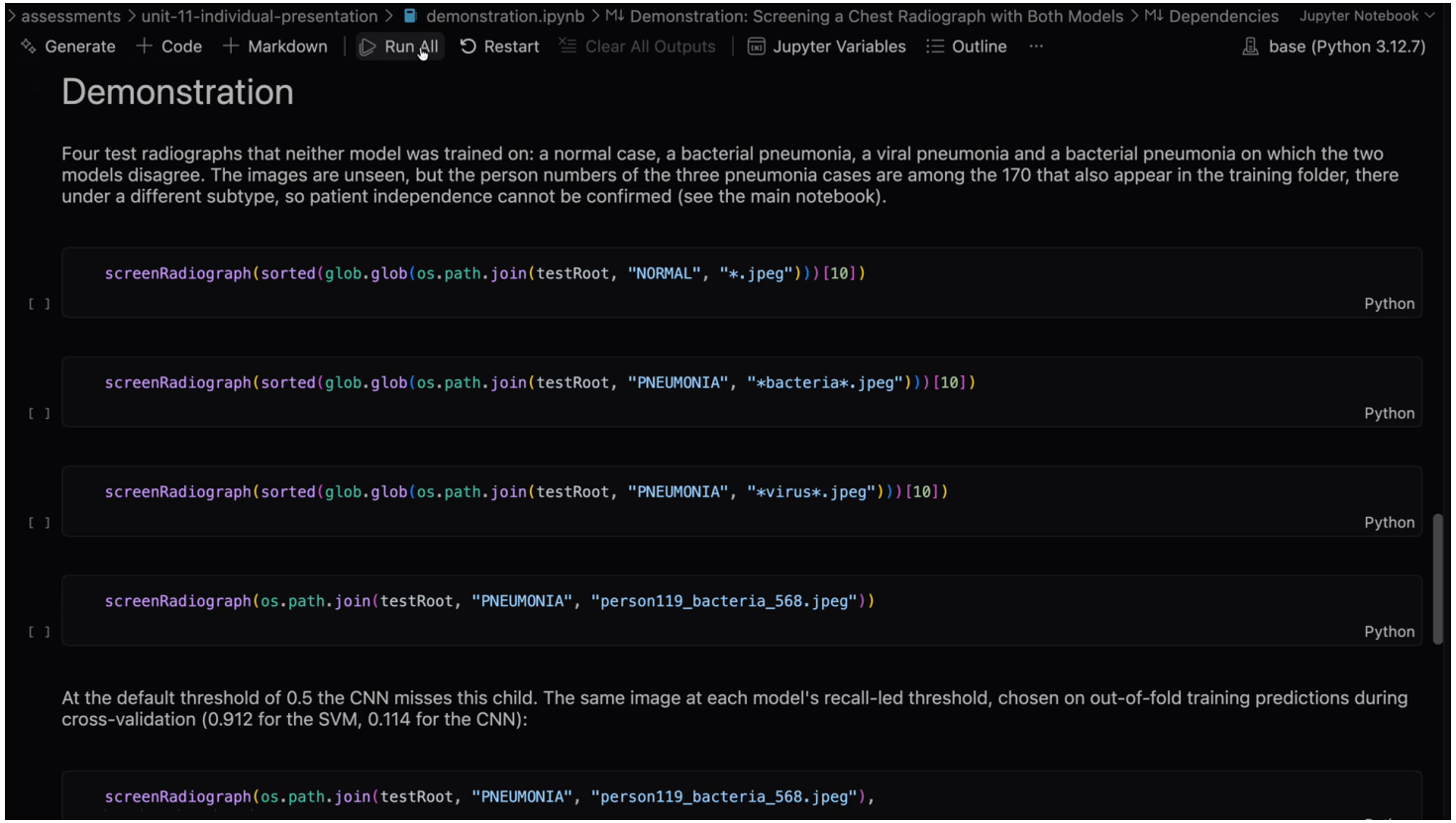


The model triages; it never reports alone

Better held-out generalisation | less shortcut-prone | Grad-CAM rationale | 14.8 ms against 2.1 ms: negligible

Comprehensive live monitoring and stale thresholds: Sculley et al. (2015)

Demonstration: screening unseen radiographs



Bacterial case	p(pneumonia)
HOG + SVM	1.000 flagged
CNN at 0.5	0.254 missed
CNN at 0.114	0.254 flagged

Same image, same score: the threshold decides whether this child is flagged

Conclusion

Validation

Cross-validation favoured the SVM; the held-out test set favoured the CNN

Explainability

Grad-CAM central, not always clinical; LIME contradicts it

Bias

A no-pixel model beat the SVM on test (0.83 against 0.78): a collection confound

Deployment

CNN as a monitored triage aid, validated on unseen hospitals first

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